Polish-Japanese Academy of Information Technology

Bachelor's engineering thesis

Short-Term Forecasting of Martian Surface Weather Using Machine Learning

PRO3, Task 01: Chapter 1, Introduction

Uladzislau Staliarou (s29905)

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1 Introduction

1.1 Background and context

On Earth, data-driven methods now rival numerical weather prediction. Machine learning (ML) models trained on about four decades of global reanalysis data, such as GraphCast and Pangu-Weather, match or exceed the operational deterministic forecasts of the European Centre for Medium-Range Weather Forecasts on most verification targets[1]. Such data volumes exist for only one planet.

Mars offers a very different setting. Its thin carbon dioxide atmosphere, with a mean surface pressure below one percent of the terrestrial value, stores little heat, so the near-surface air temperature changes by several tens of kelvin between night and day and the pressure field carries strong thermal tides[2]. Seasonally, part of the atmosphere condenses onto the winter polar cap; the Viking landers recorded a peak-to-peak swing of the daily mean pressure of 26 % of its mean value[3]. Outside the dusty season, the weather at a given site repeats closely from one sol (a Martian day of about 24 h 40 min) to the next[4].

Surface weather is measured directly by a few landed stations. Since 2012, three of them have produced long public records (Table 1.1): the Rover Environmental Monitoring Station (REMS) on Curiosity[5], the Auxiliary Payload Sensor Suite (APSS) on InSight[6] and the Mars Environmental Dynamics Analyzer (MEDA) on Perseverance[7]. The MEDA release used in this thesis, distributed by the NASA Planetary Data System (PDS), spans 1,619 sols, from February 2021 to September 2025[8].

[1] R. Lam, A. Sanchez-Gonzalez, M. Willson, et al. "Learning skillful medium-range global weather forecasting". In: *Science* 382.6677 (2023), pp. 1416–1421; K. Bi et al. "Accurate medium-range global weather forecasting with 3D neural networks". In: *Nature* 619 (2023), pp. 533–538.

[2] Cf. R. M. Haberle et al., eds. *The Atmosphere and Climate of Mars*. Cambridge Planetary Science 18. Cambridge: Cambridge University Press, 2017.

[3] S. L. Hess, R. M. Henry, and J. E. Tillman. "The seasonal variation of atmospheric pressure on Mars as affected by the south polar cap". In: *Journal of Geophysical Research* 84.B6 (1979), pp. 2923–2927.

[4] Cf. Haberle et al., *op. cit.*

[5] J. Gómez-Elvira et al. "REMS: The Environmental Sensor Suite for the Mars Science Laboratory Rover". In: *Space Science Reviews* 170 (2012), pp. 583–640.

[6] D. Banfield et al. "InSight Auxiliary Payload Sensor Suite (APSS)". in: *Space Science Reviews* 215, 4 (2019).

[7] J. A. Rodriguez-Manfredi et al. "The Mars Environmental Dynamics Analyzer, MEDA. A Suite of Environmental Sensors for the Mars 2020 Mission". In: *Space Science Reviews* 217, 48 (2021).

[8] J. A. Rodriguez-Manfredi et al. *The Mars Environmental Dynamics Analyzer, MEDA*. Data bundles of the Mars 2020 mission. NASA Planetary Data System, Atmospheres Node. 2021. URL: `https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/PERSEVERANCE/meda.html` (accessed 2026-10-07). DOI: 10.17189/1522849.

Table 1.1: In situ meteorological stations used in this thesis

Station	Platform	Landing site	Mission period	Main measured quantities
REMS	Curiosity rover	Gale Crater	since Aug 2012	air and ground temperature, pressure, wind, humidity, UV
APSS	InSight lander	Elysium Planitia	Nov 2018 – Dec 2022	air temperature, pressure, wind
MEDA	Perseverance rover	Jezero Crater	since Feb 2021	air and ground temperature, pressure, wind, humidity, radiation, thermal infrared

Source: own elaboration based on the instrument descriptions[9].

Forecasts for Martian missions currently come from physical models. General circulation models (GCMs)[10] simulate the global atmosphere on grids far coarser than a landing site, and multi-model simulations were used to predict the meteorology of Jezero before Perseverance landed[11]. Their output describes expected conditions for a season and time of sol rather than a forecast conditioned on the latest station measurements.

1.2 Motivation and problem statement

Short-term knowledge of surface conditions has direct engineering value: rover operations are planned in daily cycles, and their energy and thermal budgets depend on temperature and dust. Weather can also end a mission. The solar-powered Opportunity rover stopped communicating in June 2018, when a planet-wide dust storm cut off its sunlight[12]. A forecaster driven by a station's own recent measurements would be cheap to run and could complement physical models at one site.

Despite long station records, data-driven forecasting of Martian weather has received little attention. The closest precedent trained neural networks on REMS and, later, Viking data to predict the next sol of air temperature and pressure, reporting correlation coefficients and relative errors[13]. Neither measure shows skill over a reference forecast, and on Mars the difference

[9] Gómez-Elvira et al., *op. cit.*; Banfield et al., *op. cit.*; Rodriguez-Manfredi et al., "The Mars Environmental Dynamics Analyzer, MEDA. A Suite of Environmental Sensors".

[10] F. Forget, F. Hourdin, R. Fournier, et al. "Improved general circulation models of the Martian atmosphere from the surface to above 80 km". In: *Journal of Geophysical Research: Planets* 104.E10 (1999), pp. 24155–24175.

[11] C. E. Newman et al. "Multi-model Meteorological and Aeolian Predictions for Mars 2020 and the Jezero Crater Region". In: *Space Science Reviews* 217, 20 (2021).

[12] NASA Jet Propulsion Laboratory. *NASA's Opportunity Rover Mission on Mars Comes to End*. News release. Feb. 13, 2019. URL: https://www.jpl.nasa.gov/news/nasas-opportunity-rover-mission-on-mars-comes-to-end/ (accessed 2026-10-07).

[13] A. de Cabo García et al. "MARSchine Learning: Prediction of Mars meteorological variables using artificial neural networks". In: *CPESS 6: VI Meeting of Planetary Science and Solar System Exploration*. Extended abstract. Torrejón de Ardoz: INTA, 2019. URL: https://www.inta.es/export/sites/default/.galleries/cpess6-Descarga/de-Cabo-Garcia-A_MARSchine-Learning-Prediction-of-Mars-meteorolog

matters: the diurnal cycle is so regular that persistence, which assumes that conditions one sol ahead equal those observed now, is already accurate, and a model can correlate almost perfectly with the observations while adding little to it. Skill must therefore be measured against simple reference forecasts[14].

Two further questions remain open: whether secondary measurements, such as humidity and radiation, improve the forecasts, and whether a model trained at one landing site works at another. Because each site has its own local circulation (at Jezero, winds are driven mainly by local topography[15]), the transfer error indicates how much of the learned behavior is local.

The problem addressed in this thesis is therefore to forecast near-surface air temperature, pressure and horizontal wind speed 1 to 48 hours ahead from the recent in situ measurements of a landing site, and to establish whether ML models show measurable skill over simple reference forecasts, both at the training site and at other sites. The contribution is methodological rather than algorithmic: established models are evaluated under the practice standard in terrestrial forecasting (chronological splits, reference forecasts and skill scores), and cross-site transfer is measured rather than assumed.

1.3 Aim and objectives

The main aim of this thesis is to design, implement and evaluate a reproducible ML pipeline for short-term forecasting of Martian surface weather from in situ measurements, and to determine for which variables and horizons it outperforms simple reference forecasts. The objectives are:

- **O1** To build a reproducible pipeline that ingests the public MEDA archive, applies quality control and aggregates the data onto a 5-minute grid.
- **O2** To implement two reference forecasts: persistence and a time-of-sol climatology (seasonal-naive forecast).
- **O3** To implement two ML models: gradient-boosted decision trees on engineered features and a temporal convolutional network on a 24-hour input window.
- **O4** To evaluate all models on a chronologically held-out test period at horizons of 1, 6, 24 and 48 hours, using the root mean square error (RMSE), the mean absolute error (MAE) and skill relative to persistence.
- **O5** To quantify the effect of secondary signals (humidity, solar radiation, thermal infrared fluxes) on the primary forecasts.

`ical-variables-using-artificial-neural-networks.pdf` (accessed 2026-10-07); A. de-Cabo-García et al. "Predicción de variables meteorológicas de Marte mediante redes neuronales artificiales". In: *DYNA New Technologies* 9.1 (2022). In Spanish.

[14] Cf. A. H. Murphy. "Skill Scores Based on the Mean Square Error and Their Relationships to the Correlation Coefficient". In: *Monthly Weather Review* 116.12 (1988), pp. 2417–2424; R. J. Hyndman and G. Athanasopoulos. *Forecasting: Principles and Practice*. 3rd ed. Melbourne: OTexts, 2021. URL: `https://otexts.com/fpp3/` (accessed 2026-10-07).

[15] J. A. Rodriguez-Manfredi et al. "The diverse meteorology of Jezero crater over the first 250 sols of Perseverance on Mars". In: *Nature Geoscience* 16 (2023), pp. 19–28.

O6 To assess cross-site transfer by evaluating models trained at one site on the records of the other stations.

The objectives answer three research questions:

RQ1 How accurately can ML models forecast 1–48 h Martian surface weather at a single site, compared with persistence and seasonal-naive forecasts?

RQ2 Do secondary signals measurably improve the forecasts of the primary variables?

RQ3 Does a model trained at one site generalize to another, and what does the transfer error reveal about local versus global weather structure?

Objectives O1–O4 address RQ1, O5 addresses RQ2 and O6 addresses RQ3. The hypotheses and success criteria are stated in Chapter 6.

1.4 Scope of the study

The study is limited to point forecasts of air temperature, pressure and horizontal wind speed at horizons of 1 to 48 hours, using station measurements only (no orbital data or physical model output). The primary site is Jezero Crater (MEDA sols 1 to 1619, 19 February 2021 to 9 September 2025), with derived products used where they exist and calibrated products for air temperature and radiation. REMS and APSS provide the transfer sites. Derived MEDA wind covers only sols 36 to 312, so wind is evaluated on that window alone. Instrument-team quality control is accepted as released.

Out of scope are probabilistic forecasting, the forecasting of dust storms or dust devils, data assimilation into physical models, the vertical structure of the atmosphere and operation on flight hardware. All experiments were designed for a single consumer laptop with a 4 GB graphics processing unit (GPU), which bounds the model size. The software is a research pipeline; Chapter 5 outlines how it could be deployed as a forecasting service.

1.5 Thesis structure

Chapter 2 reviews the literature on Martian meteorology, ML on Mars data and data-driven forecasting. Chapter 3 documents the data sources, formats, folder structure and licensing. Chapter 4 presents the exploratory analysis and the standardization and normalization of the data. Chapter 5 describes the architecture of the software system. Chapter 6 describes the forecasting problem, the models and the evaluation setup. Chapter 7 presents and discusses the results for each research question. Chapter 8 summarizes the work, draws conclusions and proposes future work.

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